# NETWORKWALKS-B082

## FOOTPRINTING & NETWORK SCANNING PHASES

**W2-PM-FINAL | CYBERSECURITY**


| |
|---|
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| |
|---|
| **Target:** Networkwalks (written permission secured), Own Local LAN Network |
| **Classification:** Confidential |
| **Version:** 1.0 |
| **Modules Completed**: |
| PM1-FOOTPRINT-WITH-MULTIPLE-TOOLS |
| PM2-FOOTPRINT-WITH-GHDB |
| PM3-FOOTPRINT-WITH-MALTEGO |
| PM4-FOOTPRINT-WITH-theHarvester |
| PM5-NETWORK-SCANNING-WITH-ZENMAP |

Date

June 24, 2026

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# 1. Liability Disclaimer

This report documents reconnaissance and network-scanning activities performed strictly for educational purposes under the NetworkWalks cybersecurity training program, Batch NETWORKWALKS-B084D. All activities against the external target, Networkwalks (networkwalks.com), were carried out only after securing explicit written permission from the organization. All internal network-scanning activities (Module 5 / WK2-PM5) were performed exclusively against the author's own local LAN subnet, which the author owns and administers.

**Google Hacking Database (Module 2 / WK2-PM2):**

Unlike the other modules, GHDB-based dorking queries a public search engine's index and can surface systems, devices, and files belonging to unrelated third parties who are not party to this engagement's written-permission agreement. No third-party system discovered through this technique was accessed, interacted with, downloaded from, or further probed beyond confirming its existence in search results. No live camera feed discovered was accessed, viewed, recorded, or shared. Findings from this module are retained privately and are not reproduced in this report; see the dedicated findings file referenced in Section 4.2.

**OSINT / email harvesting (Modules 3 & 4 / WK2-PM3, WK2-PM4):**

Email addresses and related data discovered via Maltego and theHarvester constitute personally identifiable information (PII) belonging to real individuals. This data has been handled in line with the scope of the written-permission agreement and applicable privacy regulations, has not been used for any outreach, phishing, or unsolicited communication, and is not reproduced in full within this report.

**General:**

This document and the techniques it describes must not be used against any system, network, or individual without separate, explicit written authorization. Unauthorized use may violate computer-misuse, data-protection, and anti-spam legislation, including but not limited to the Prevention of Electronic Crimes Act (PECA) 2016 (Pakistan), the Computer Fraud and Abuse Act (CFAA) (United States), the Computer Misuse Act 1990 (United Kingdom), and the General Data Protection Regulation (GDPR) (European Union), depending on jurisdiction. The author and

NetworkWalks accept no responsibility for any misuse of the information, methodology, or findings contained in this document.

## 2. Introduction

This report consolidates five training modules completed during Week 2 of the NetworkWalks cybersecurity program into a single, structured penetration-testing-style report. Together, the modules cover the Footprinting & Reconnaissance and Network Scanning phases that open most real-world penetration testing engagements and attacker kill-chains alike — the stage at which the greatest volume of information about a target is gathered with the least amount of direct interaction and risk.

Reconnaissance activities are generally classified as passive or active. Passive reconnaissance collects information without directly touching the target's own infrastructure — querying public registries, search engines, OSINT platforms, and third-party indexes — and is therefore very difficult for a target to detect. Active reconnaissance sends traffic directly to the target's systems (web requests, DNS queries, network probes) and, while still non-destructive, can appear in logs and trigger security alerts.

The five modules documented in this report map to these categories as follows:

- PM1 — Footprinting with Multiple Kali Tools: a mixed passive/active assessment of one authorized external target (networkwalks.com) using whois, whatweb, nslookup, curl, wafw00f, and dnsrecon.
- PM2 — Footprinting with GHDB: passive, internet-wide OSINT using Google Dorking and the Google Hacking Database.
- PM3 — Footprinting and Email Harvesting using Maltego: passive OSINT and link-analysis to enumerate an organization's public email footprint.
- PM4 — Footprinting with theHarvester: passive OSINT comparing single-source versus multi-source email/subdomain enumeration.
- PM5 — Network Scanning with Zenmap: active internal host discovery and topology mapping against the author's own LAN.

The engagement therefore spans two distinct scopes: an authorized external target (Networkwalks) and the author's own internal network, each handled under the authorization and handling terms set out in Section 1.

## 3. Tools Used

The following tools and techniques were used across the five modules:

| Module | Tool / Technique | Category | Recon Type |
|---|---|---|---|
| Platform | Kali Linux | Operating System | — |
| PM1 | whois | Domain Intelligence | Passive |
| PM1 | whatweb | Web Fingerprinting | Active |
| PM1 | nslookup | DNS Resolution | Passive |
| PM1 | curl -I | HTTP Client | Active |
| PM1 | wafw00f | WAF Detection | Active |
| PM1 | dnsrecon | DNS Enumeration | Active |
| PM2 | Google Search (Advanced Operators) | Passive OSINT | Passive |
| PM2 | Google Hacking Database (GHDB) | Dork Repository | Passive |
| PM3 | Maltego (Community Edition) | OSINT / Link Analysis | Passive |
| PM4 | theHarvester | OSINT / Email & Subdomain Enumeration | Passive |
| PM5 | Zenmap | Network Scanning (GUI) | Active |
| PM5 | Nmap | Network Scanning (Engine) | Active |

# 4. Activities Performed

## 4.1 Footprinting & Reconnaissance with Multiple Kali Tools

| Field | Result |
|---|---|
| Target | networkwalks.com (written permission secured) |
| Objective | Establish a baseline domain, DNS, web-technology, and HTTP security-header footprint using six complementary command-line tools. |

**Key Findings:**

- whois networkwalks.com — domain registration details (registrar, creation/expiry dates, name servers) recorded; see WK2-PM1 README for full detail.
- whatweb networkwalks.com — Apache web server (Nginx caching layer indicated), WordPress 7.1.4 with WordPress Download Manager plugin 3.3.76, jQuery 3.7.1, Google Tag Manager; a contact email (info@networkwalks.com) and a WordPress generator version tag were both publicly exposed.
- nslookup networkwalks.com — domain resolved to IP 192.232.216.135 (United States); confirmed consistent with the WhatWeb and curl results.
- curl -I https://networkwalks.com — HTTP/2 200 OK; Server header discloses “Apache” with no version number. permissions-policy and referrer-policy headers present; Strict-Transport-Security, Content-Security-Policy, X-Frame-Options, and X-Content-Type-Options headers were all absent. The __wpdm_client cookie was set with Secure and HttpOnly flags but no SameSite attribute. The WordPress REST API endpoint (/wp-json/) was exposed via the Link header.
- wafw00f https://networkwalks.com — WAF detection result recorded; see WK2-PM1 README / WAFW00F-TOOL-RESULT for the specific vendor finding.
- dnsrecon -d networkwalks.com — full DNS record enumeration (A, MX, NS, TXT, SOA) recorded; see WK2-PM1 README / DNSRECON-TOOL-RESULT for detail.

Full raw tool output and screenshot evidence are catalogued in the WK2-PM1 README and referenced in Section 7.1 of this report.

## 4.2 Footprinting with GHDB

| Field | Result |
|---|---|
| Target | Internet-wide (unrelated third-party systems — see Section 1 scope note) |
| Objective | Use Google Dorking and the GHDB to identify 10 internet-exposed security camera interfaces and 10 downloadable mathematics eBook (PDF) listings, recording the dork used for each result. |

**Key Findings:**

Both result sets were compiled using operator-based dorks (intitle:, inurl:, intext:, filetype:) drawn from the GHDB's categorized query library. In line with the scope note in Section 1, the full tables of live links and their corresponding dorks are withheld from this report and retained privately in the dedicated findings file for data-handling reasons, rather than reproduced here.

- Task 1 — Security cameras: 10 live, internet-exposed camera interfaces located via GHDB “Various Online Devices” category dorks; no feed was accessed or viewed beyond confirming reachability.
- Task 2 — Mathematics eBooks: 10 open directory/file listings containing downloadable PDF mathematics eBooks located via filetype:pdf-based dorks.

Full results are catalogued in Findings-W2-M2.docx (private) and referenced in Section 7.2 of this report.

## 4.3 Footprinting and Email Harvesting using Maltego

| Field | Result |
|---|---|
| Target | networkwalks.com (written permission secured) |
| Objective | Enumerate publicly discoverable email addresses related to the target organization's domain using Maltego Transforms. |

Key Findings:

- A Maltego Community Edition (CE) account was created and verified to enable access to the Transform Hub.
- A domain-to-email Transform was executed against the target domain, returning a set of discoverable email addresses visualized on the entity graph.
- The resulting addresses and any naming convention identified are recorded in the WK2-PM3 README; full detail is withheld from this report in line with the PII handling note in Section 1.

Full evidence (account setup through to results) is catalogued in the WK2-PM3 README and referenced in Section 7.3 of this report.

## 4.4 Footprinting with theHarvester

| Field | Result |
|---|---|
| Target | microsoft.com |
| Objective | Compare email/subdomain enumeration coverage between a single data source and an aggregated, multi-source run. |

**Key Findings:**

- Task 1 — theHarvester -d microsoft.com -b baidu -l 1000: results obtained from the Baidu source alone, with the result limit set to 1000.
- Task 2 — theHarvester -d microsoft.com -b all -l 50: results obtained by aggregating all configured sources, capped at a limit of 50.
- Comparing the two runs showed that source breadth affects completeness independently of result limit — a lower-limit, all-source run can surface subdomains and emails that a single high-limit source run does not, and vice versa; see the WK2-PM4 README for the specific comparison.

Full compiled lists of discovered email addresses and subdomains are retained privately in Findings-W2-M4.docx and referenced in Section 7.4 of this report.

## 4.5 Network Scanning with Zenmap

| Field | Result |
|---|---|
| Target | Own local LAN subnet — 10.0.0.0/24 |
| Objective | Perform host discovery across the local subnet, identify live hosts' IP and MAC addresses, and visualize the network topology. |

**Key Findings:**

| Field | Result |
|---|---|
| Live hosts identified | 4 |
| IP addresses | 10.0.0.1, 10.0.0.2, 10.0.0.9, 10.0.0.11 |
| MAC addresses | 52:55:0A:00:00:01, 08:00:27:DC:12:97, 08:00:27:35:A9:61 |
| Topology output | Exported and saved as Topology-Graphics.pdf on the desktop |

Note: only three MAC addresses were returned for four live hosts. This is expected ARP-based host-discovery behavior — the scanning machine itself does not resolve a MAC address for its own IP via ARP, since ARP only applies to other devices on the segment.

Full evidence is catalogued in the WK2-PM5 README and referenced in Section 7.5 of this report.

## 5. Recommendations

**Domain & DNS Hygiene**

- Enable WHOIS privacy protection and use role-based contact email addresses rather than personal ones.
- Publish and enforce strict SPF, DKIM, and DMARC records to reduce email-spoofing risk.
- Remove stale or unused DNS records and restrict zone transfers to authorized secondary name servers only.

**Web Application Hardening**

- Suppress version banners (Server header, X-Powered-By, CMS generator meta tags) to reduce version-targeted exploitation risk.
- Implement the missing security headers identified in Module 1: Strict-Transport-Security (HSTS), Content-Security-Policy (CSP), X-Frame-Options, and X-Content-Type-Options.
- Add a SameSite attribute to all session and application cookies.
- Restrict or authenticate access to the WordPress REST API (/wp-json/) where public access is not required.
- Keep WordPress core and all plugins (e.g., WordPress Download Manager) patched to their latest versions.

**WAF & Perimeter Defense**

- Deploy and keep a Web Application Firewall updated and tuned; combine with request rate limiting.
- Never rely on a WAF as the sole control — pair it with regular patching and secure coding practices.

**OSINT & Human-Layer Exposure**

- Minimize the volume of employee email addresses published publicly, and avoid a predictable, easily-guessable naming convention.
- Conduct regular phishing-awareness training for staff, reinforced with simulated phishing exercises under separate written authorization.
- Periodically monitor certificate-transparency logs and search-engine indexes for unexpected organizational exposure.

**Internal Network Security**

- Segment the network (VLANs, inter-segment firewall rules) to limit what a single compromised host can reach.
- Disable unnecessary ICMP/ARP responses on sensitive hosts where operationally feasible.

- Deploy monitoring (IDS/IPS) to detect unexpected internal host-discovery or port-scanning traffic.

**General**

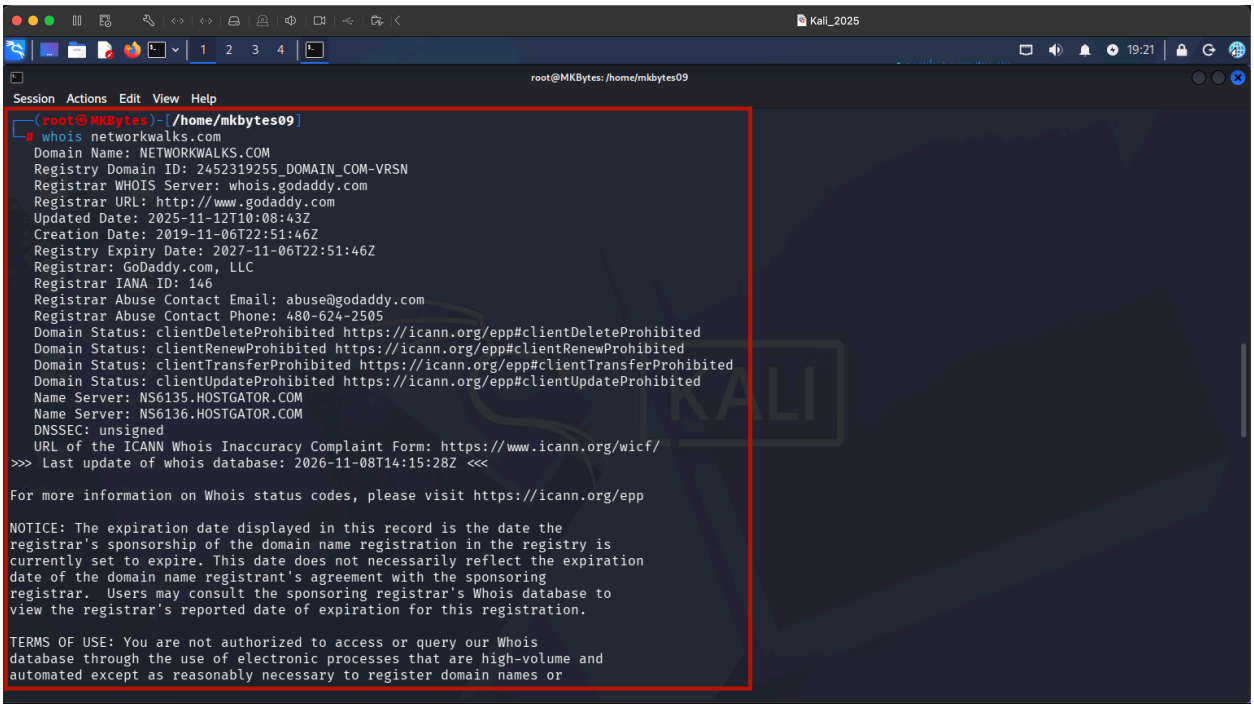
- Repeat this reconnaissance exercise periodically against the organization's own assets, so new exposure is identified internally before an external attacker finds it.

## 7. Evidence Collection

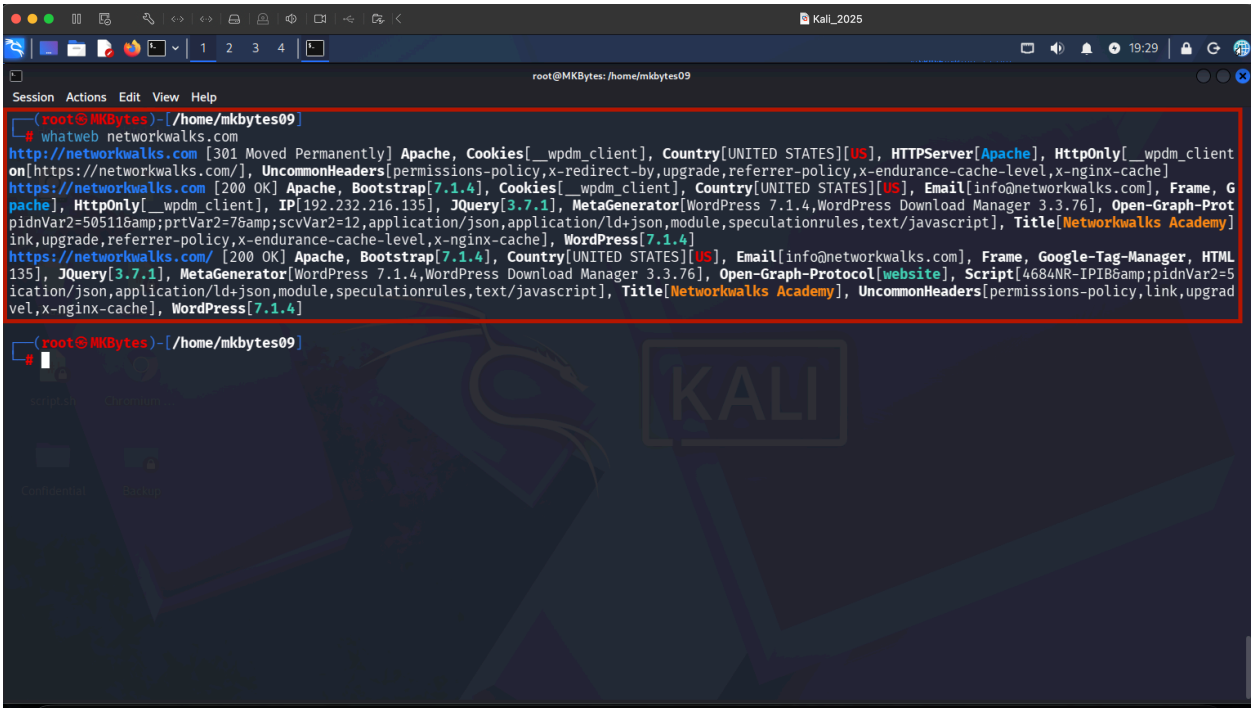
This section is intentionally left as a placeholder. Insert the corresponding screenshots and exported files under each module heading below before final submission.

### 7.1 WK2-PM1 — Footprinting with Multiple Kali Tools

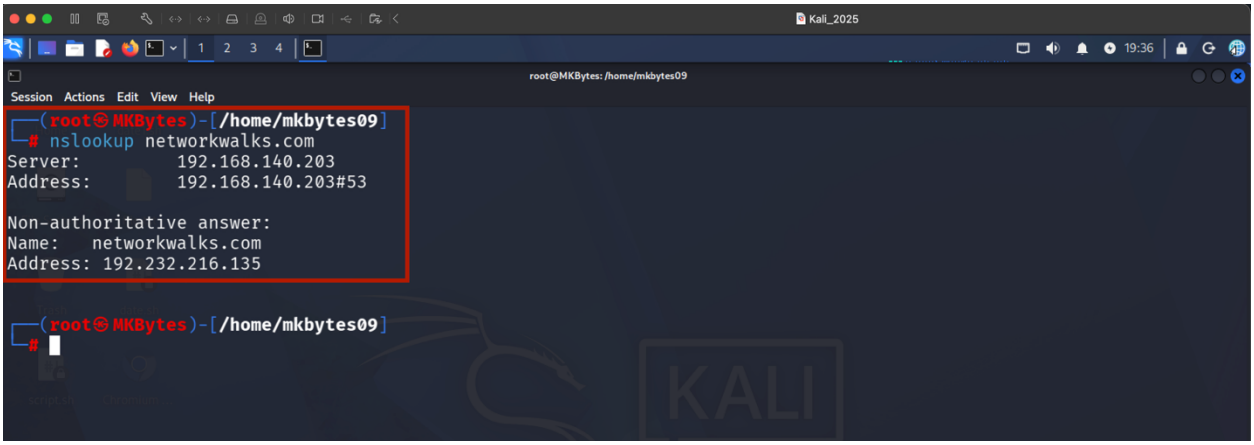
**WHOIS-TOOL-RESULT:**



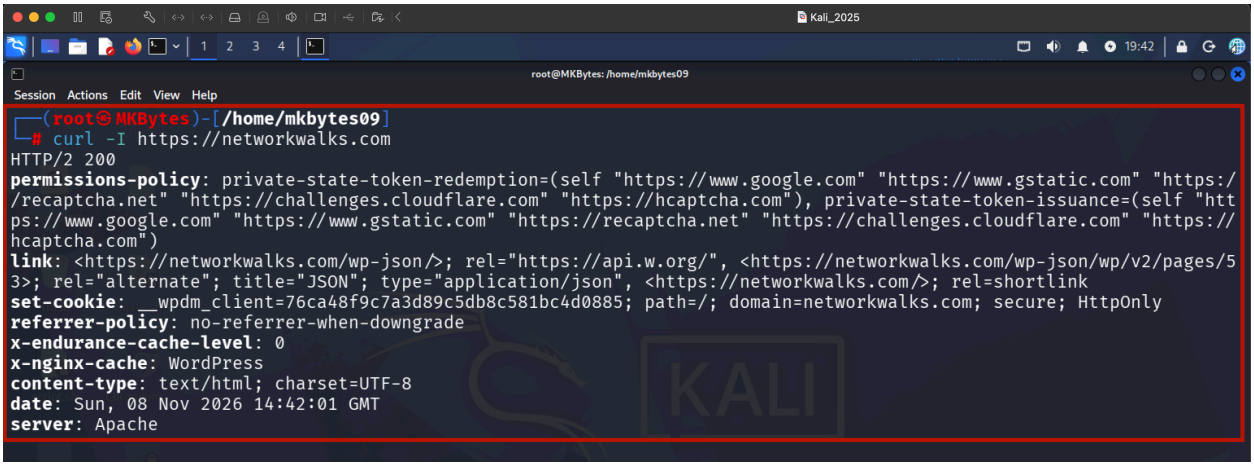
## WHATWEB-TOOL-RESULT:



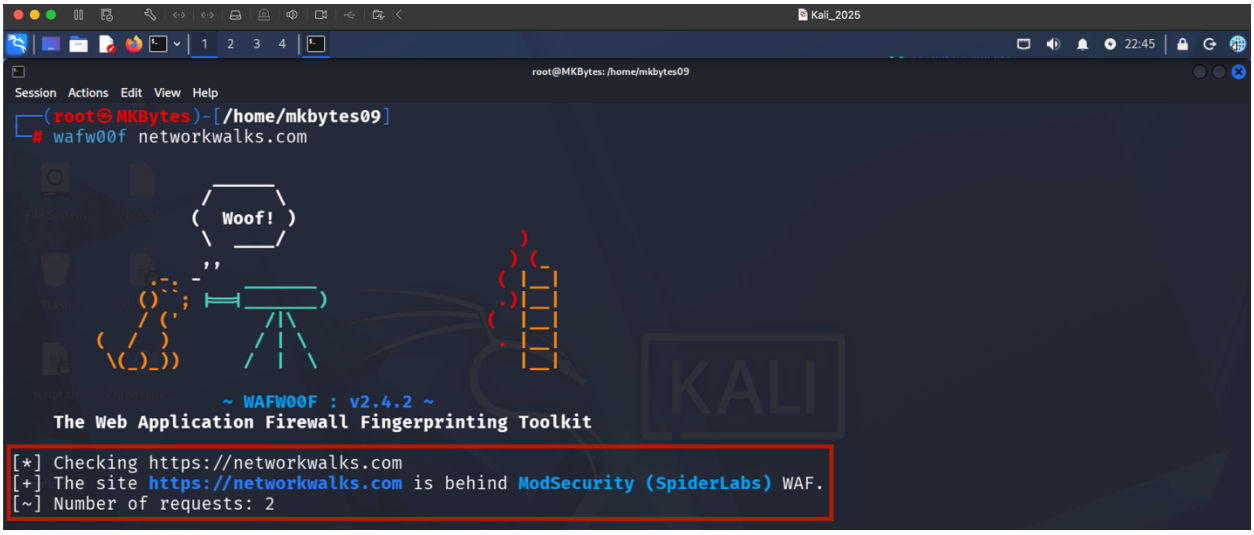

## NSLOOKUP-TOOL-RESULT:



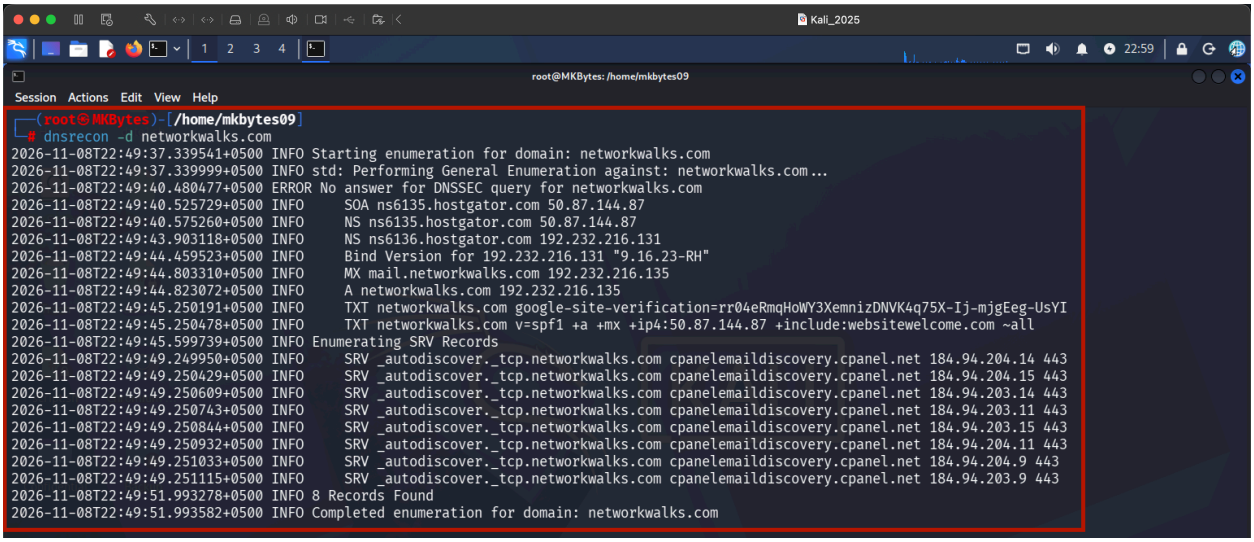
## CURL-TOOL-RESULT:




## WAFW00F-TOOL-RESULT:



**DNSRECON-TOOL-RESULT:**

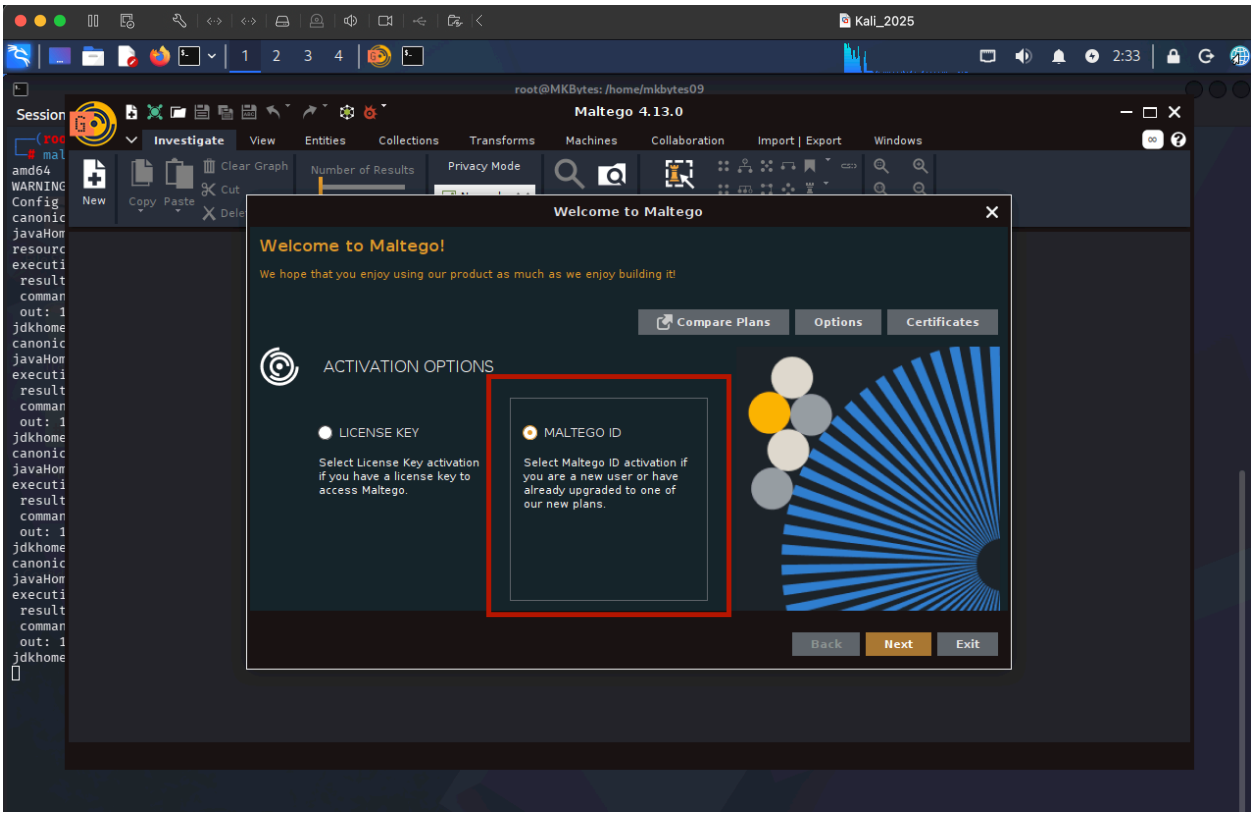



## 7.2 WK2-PM2 — Footprinting with GHDB

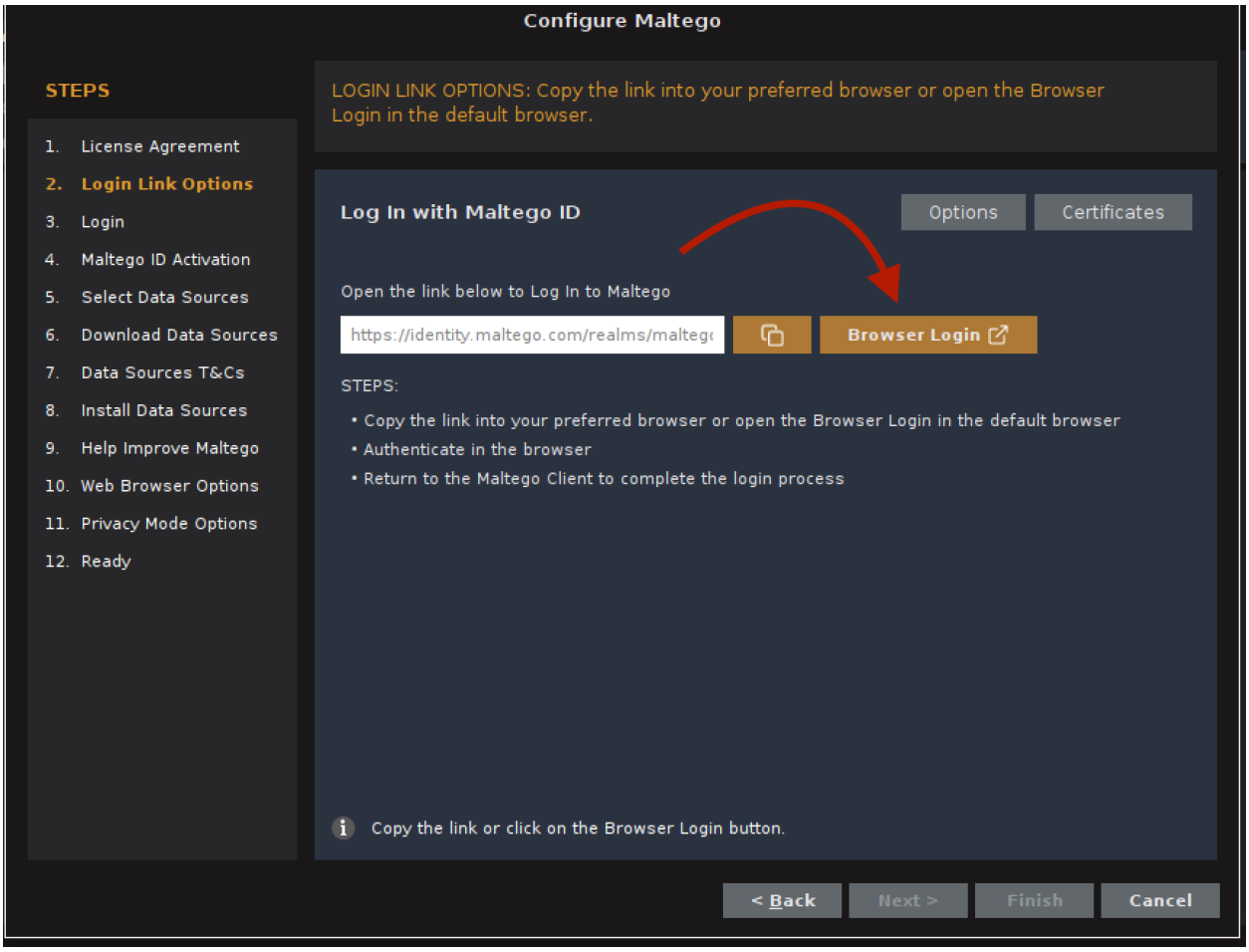
- Findings-W2-M2.docx (private — not for public distribution)

## 7.3 WK2-PM3 — Footprinting and Email Harvesting using Maltego

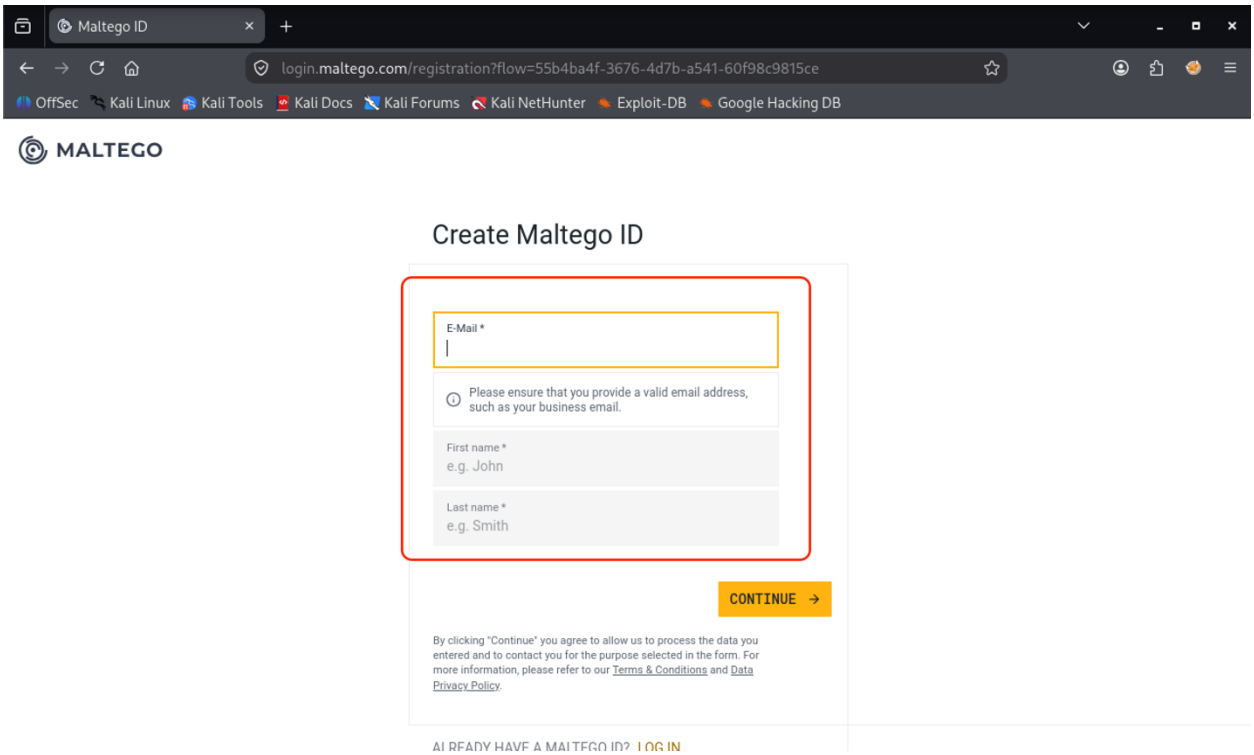
**Maltego-ID:**



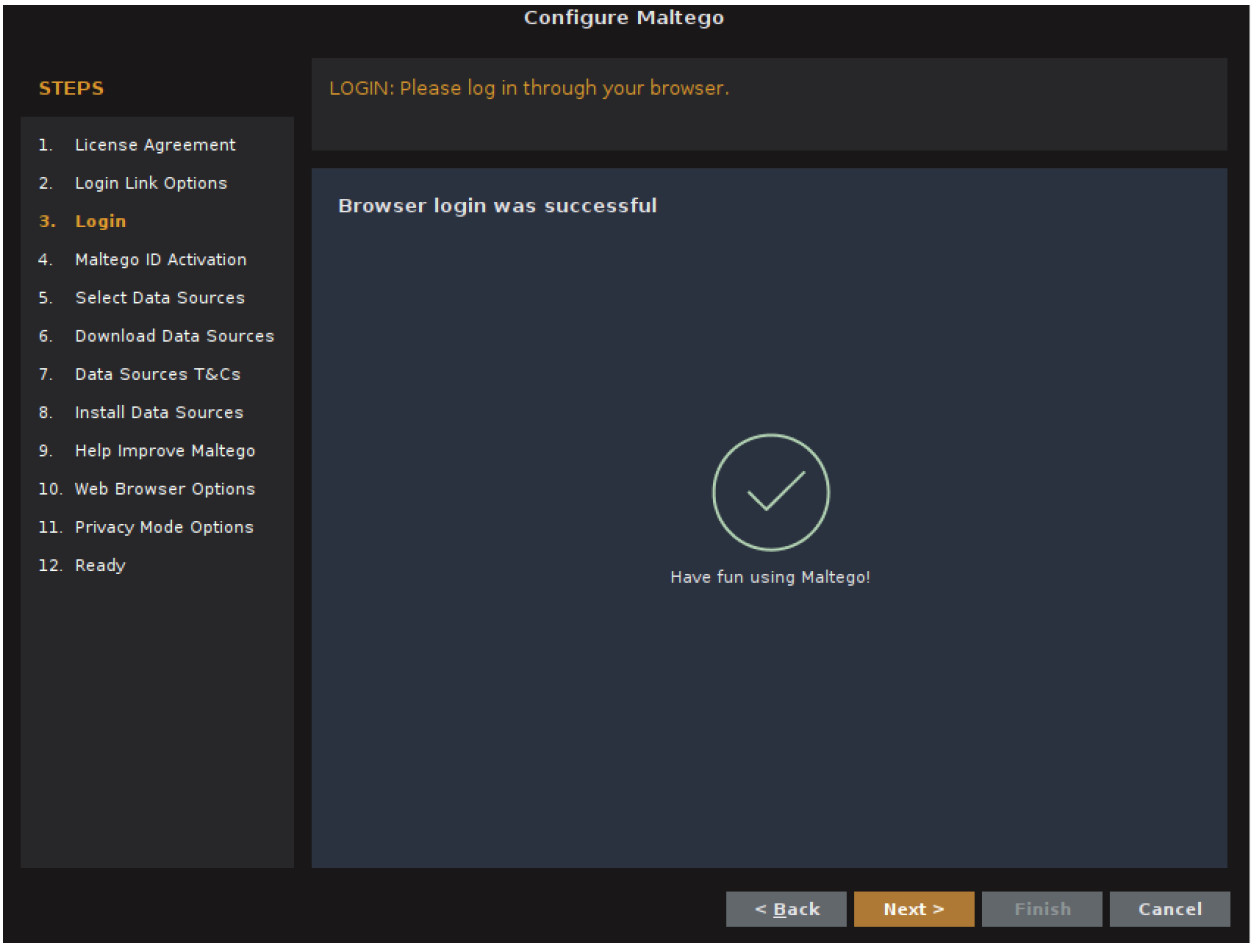

**Login-Link-Options:**



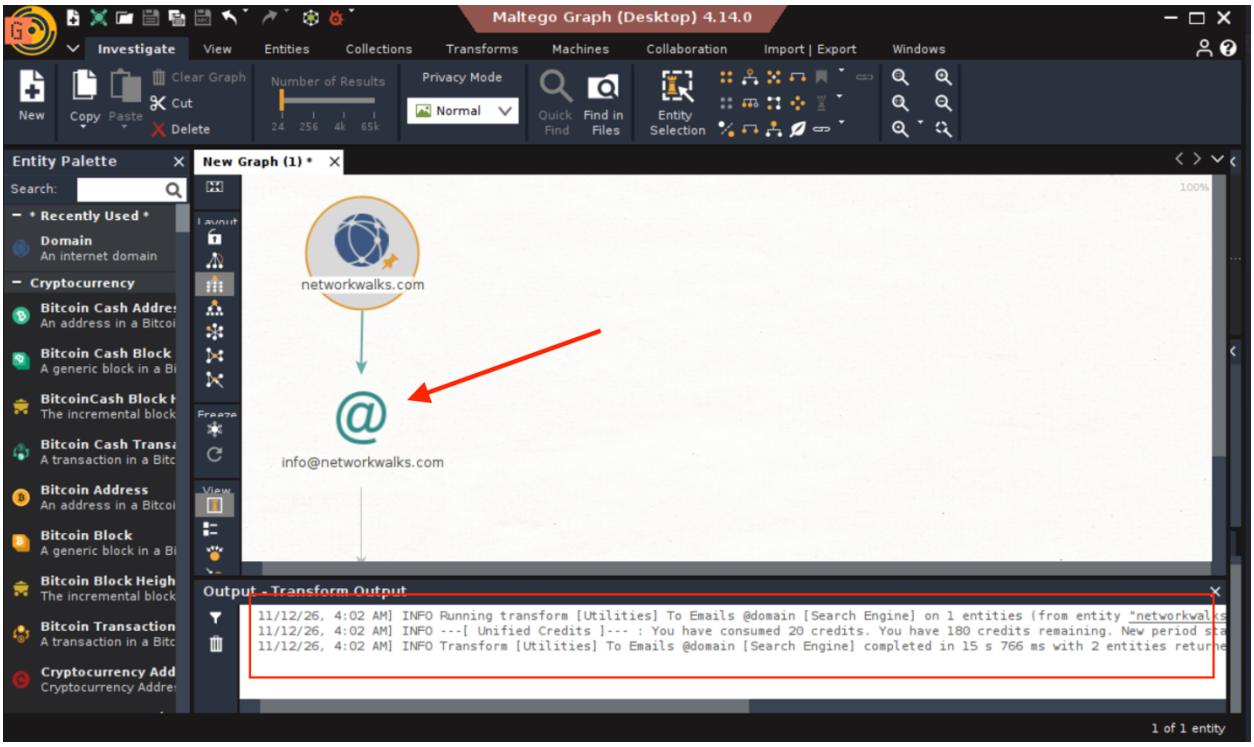

**Create-Maltego-ID:**




**Account-Verified:**

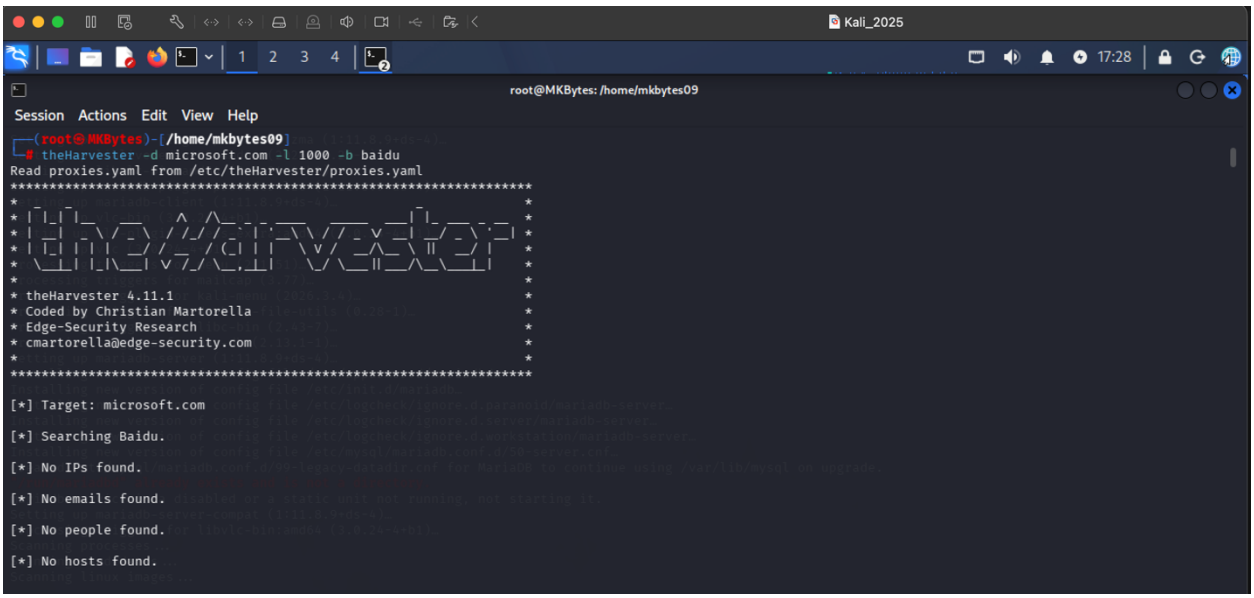



**Finding-Email-Address:**

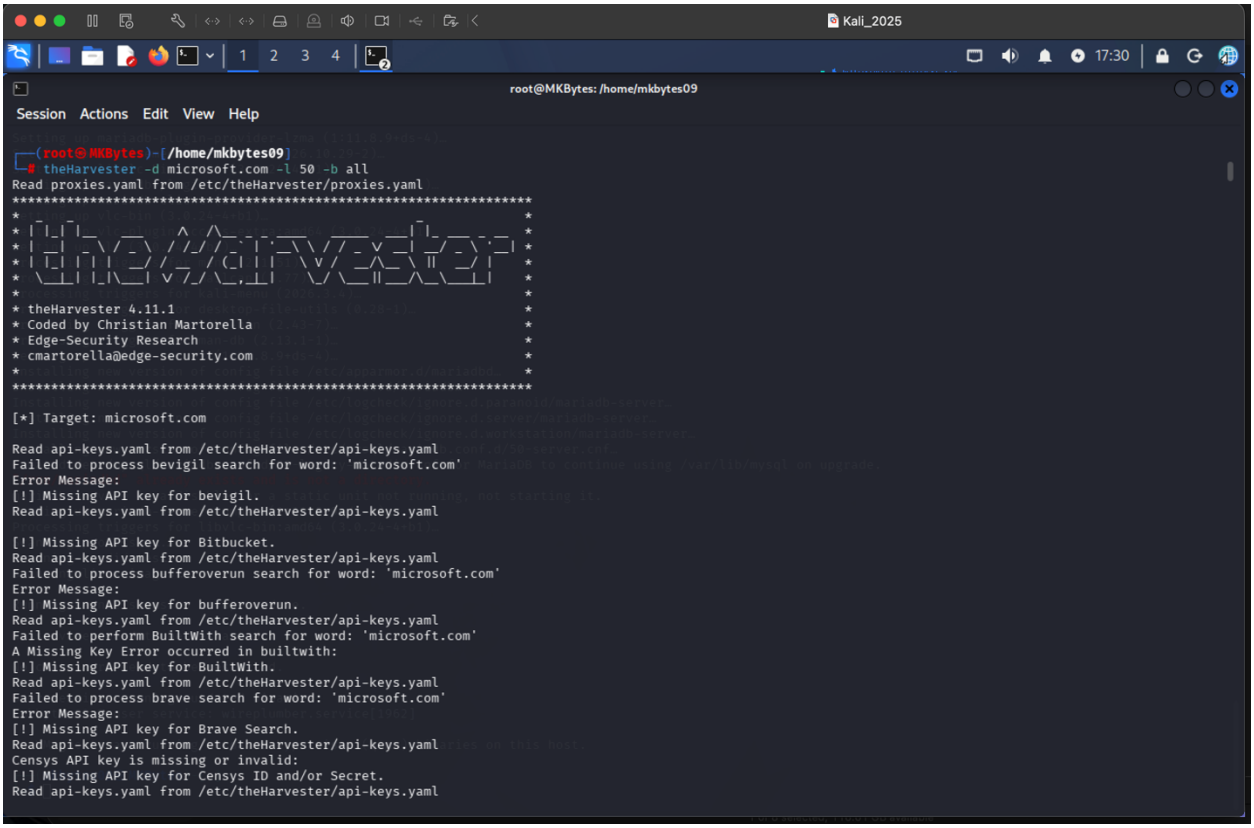



## 7.4 WK2-PM4 — Footprinting with theHarvester

**theHarvester-Result1:**



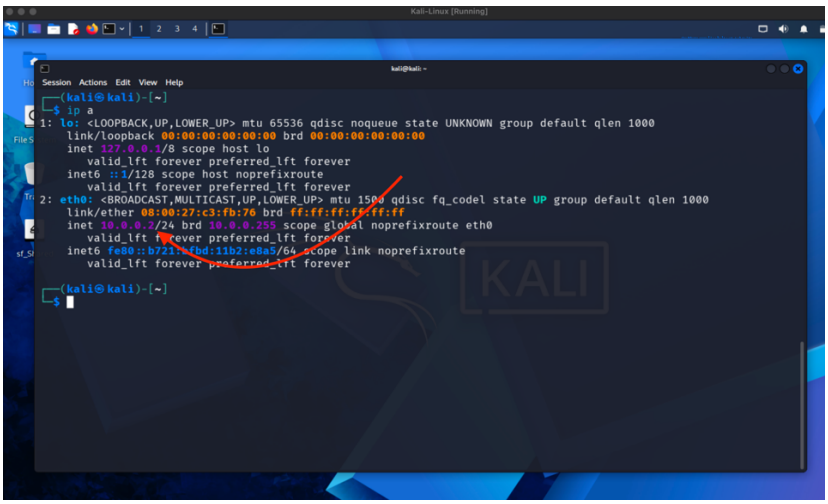
**theHarvester-Result2:**



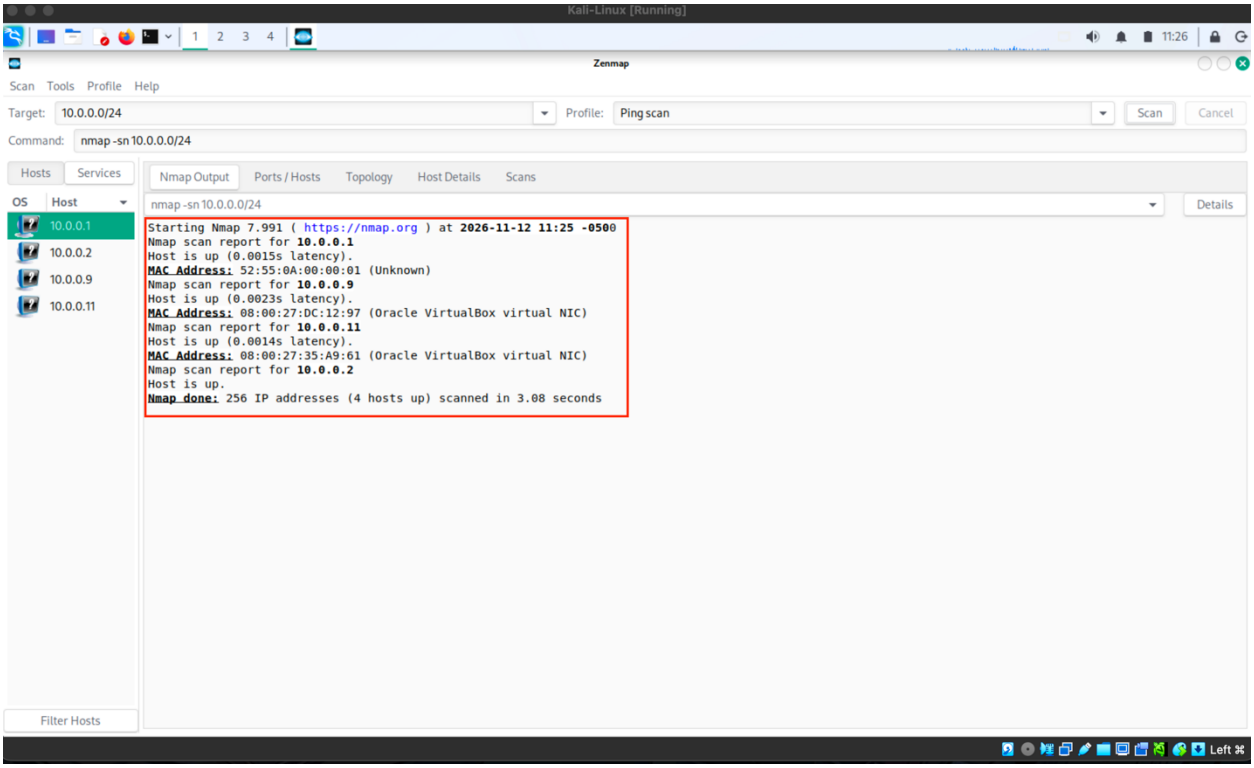

**Findings-W2-M4.docx (private — not for public distribution)**

## 7.5 WK2-PM5 — Network Scanning with Zenmap

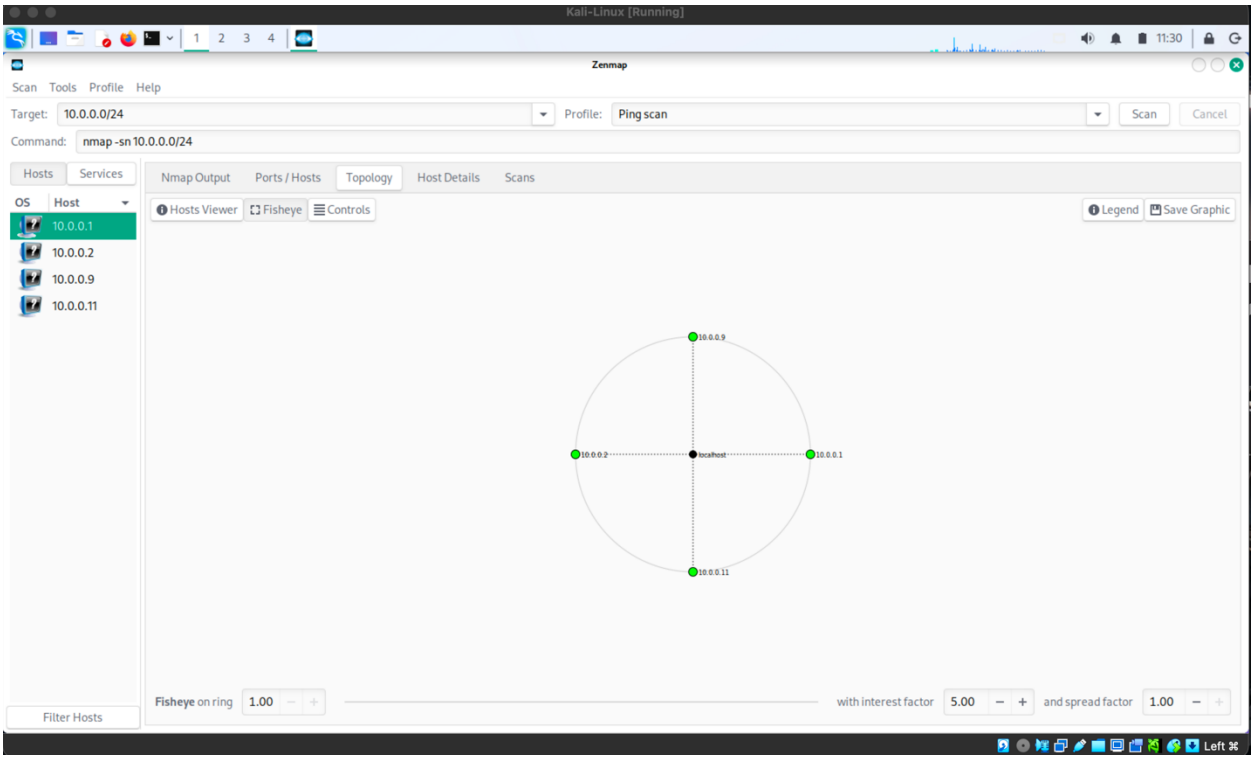
**Local-Network-Details:**



## Live-Hosts:




## Topology-Graph:



## 6. Conclusion

Across five modules, this engagement demonstrated that a substantial amount of an organization's attack surface — domain registration data, web-technology versions, missing HTTP security headers, DNS configuration, publicly indexed devices and files, employee email addresses, and subdomains — can be mapped using freely available tools and public data sources, almost entirely without ever triggering a security alert on the target's own systems. The final module then showed the other side of reconnaissance: once any foothold exists inside a network, even a simple host-discovery scan reveals exactly what else is reachable from that point.

The consistent theme across every module is that exposure is rarely the result of a sophisticated attack technique — it is overwhelmingly the result of default configurations, unnecessary information disclosure, and a lack of network segmentation. The recommendations in Section 5 are intentionally practical and low-cost relative to the exposure they close, reflecting the fact that most of what was found here could be remediated through configuration changes rather than new tooling or budget.

This report, together with its five source modules, satisfies the Week 2 Project requirements of the NetworkWalks cybersecurity training program (Batch NETWORKWALKS-B084D) and establishes the reconnaissance baseline that would inform the next phase of a real-world engagement: vulnerability analysis and, where separately authorized, controlled exploitation.

**THE END**